Ayush Kumar

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Test Date: February 24, 2026

Computer Science	Logical Ability	Computer Programming	Quantitative Ability (Advanced)
47 /100	39 /100	46 /100	74 /100

English Comprehension	WriteX - Essay Writing	Automata	Automata Fix
63 /100	75 /100	90 /100	58 /100

Personality
Completed

Computer Science — 47 / 100

OS and Computer Architecture	DBMS	Computer Networks
32 / 100	71 / 100	59 / 100

Logical Ability — 39 / 100

Inductive Reasoning	Deductive Reasoning	Abductive Reasoning
35 / 100	37 / 100	44 / 100

Computer Programming — 46 / 100

Basic Programming	Data Structures	OOP and Complexity Theory
49 / 100	51 / 100	39 / 100

Quantitative Ability (Advanced) — 74 / 100

Basic Mathematics	Advanced Mathematics	Applied Mathematics
74 / 100	71 / 100	77 / 100

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English Comprehension

63 / 100 | CEFR: **B2**

Grammar	Vocabulary	Comprehension
58 / 100	65 / 100	67 / 100

WriteX - Essay Writing

75 / 100 | CEFR: **C1**

Content Score	Grammar Score
81 / 100	59 / 100

Automata

90 / 100

Programming Ability	Programming Practices	Functional Correctness
80 / 100	100 / 100	63 / 100

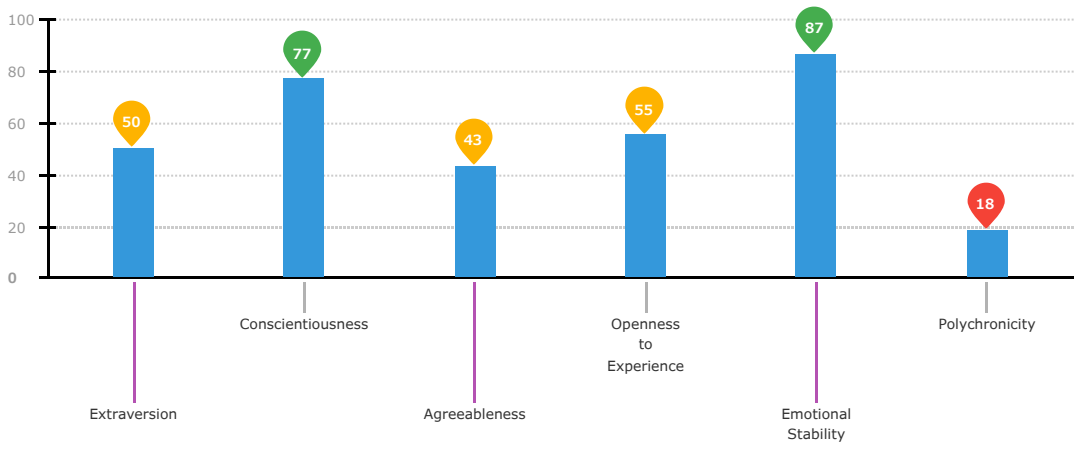
Automata Fix

58 / 100

Logical Error	Code Reuse	Syntactical Error
75 / 100	0 / 100	100 / 100

Personality

Completed



Competencies

People Interaction

Self-Drive

Trainability

Repetitive Job Suitability

Work attributes

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1 | Introduction

About the Report

This report provides a detailed analysis of the candidate's performance on different assessments. The tests for this job role were decided based on job analysis, O*Net taxonomy mapping and/or criterion validity studies. The candidate's responses to these tests help construct a profile that reflects her/his likely performance level and achievement potential in the job role

This report has the following sections:

The **Summary** section provides an overall snapshot of the candidate's performance. It includes a graphical representation of the test scores and the subsection scores.

The **Insights** section provides detailed feedback on the candidate's performance in each of the tests. The descriptive feedback includes the competency definitions, the topics covered in the test, and a note on the level of the candidate's performance.

The **Response** section captures the response provided by the candidate. This section includes only those tests that require a subjective input from the candidate and are scored based on artificial intelligence and machine learning.

The **Learning Resources** section provides online and offline resources to improve the candidate's knowledge, abilities, and skills in the different areas on which s/he was evaluated.

Score Interpretation

All the test scores are on a scale of 0-100. All the tests except personality and behavioural evaluation provide absolute scores. The personality and behavioural tests provide a norm-referenced score and hence, are percentile scores. Throughout the report, the colour codes used are as follows:

- 70 ≤ Score < 100
- 30 ≤ Score < 70
- 0 ≤ Score < 30

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2 | Insights

English Comprehension

63 / 100 | CEFR: **B2**

This test aims to measure your vocabulary, grammar and reading comprehension skills.

You have a good understanding of commonly used grammatical constructs. You are able to read and understand articles, reports and letters/mails related to your day-to-day work. The ability to read, understand and interpret business-related documents is essential in most jobs, especially the ones that involve research, technical reading and content writing.

Logical Ability

39 / 100

Inductive Reasoning

35 / 100

This competency aims to measure the your ability to synthesize information and derive conclusions.

You need to put in a great deal of effort to improve your predictive reasoning skills. Inductive reasoning will help you derive general rules from specific situations and carry out various tasks without needing instructions from others.

Deductive Reasoning

37 / 100

This competency aims to measure the your ability to synthesize information and derive conclusions.

You are able to work out simple rules based on specific evidence or information. This skill is required in high end analytics jobs where one is required to infer patterns based on predefined rules from different sets of data.

Abductive Reasoning

44 / 100

Quantitative Ability (Advanced)

74 / 100

This test aims to measure your ability to solve problems on basic arithmetic operations, probability, permutations and combinations, and other advanced concepts.

You have a strong hold on basic arithmetic and concepts of algebra. You are able to convert real-world problems into equations and solve them.

Personality

Competencies

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Extraversion

Extraversion refers to a person's inclination to prefer social interaction over spending time alone. Individuals with high levels of extraversion are perceived to be outgoing, warm and socially confident.

- You are comfortable socializing to a certain extent. You prefer small gatherings in familiar environments.
- You feel at ease interacting with your close friends but may be reserved among strangers.
- You indulge in activities involving thrill and excitement that are not too risky.
- You contemplate the consequences before expressing any opinion or taking an action.
- You take charge when the situation calls for it and you are comfortable following instructions as well.
- Your personality may be suitable for jobs demanding flexibility in terms of working well with a team as well as individually.

Conscientiousness

Conscientiousness is the tendency to be organized, hard working and responsible in one's approach to your work. Individuals with high levels of this personality trait are more likely to be ambitious and tend to be goal-oriented and focused.

- You value order and self discipline and tends to pursue ambitious endeavours.
- You believe in the importance of structure and is very well-organized.
- You carefully review facts before arriving at conclusions or making decisions based on them.
- You strictly adhere to rules and carefully consider the situation before making decisions.
- You tend to have a high level of self confidence and do not doubt your abilities.
- You generally set and work toward goals, try to exceed expectations and are likely to excel in most jobs, especially those which require careful or meticulous approach.

Agreeableness

Agreeableness refers to an individual's tendency to be cooperative with others and it defines your approach to interpersonal relationships. People with high levels of this personality trait tend to be more considerate of people around them and are more likely to work effectively in a team.

- You are flexible regarding your opinions and be willing to accommodate the needs of others.
- You are generally considerate of the needs of others yet may, at times, overlook social norms to achieve personal success.
- You are selective about the people you choose to trust.
- You are caring and you empathise a friend in distress.
- You give credit to others but also tends to be open with your friends about personal achievements.
- You are more inclined to strike a compromise in tough situations and may be suitable for jobs that demand managing expectations among different stakeholders.

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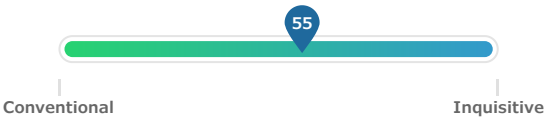
Openness to Experience



Openness to experience refers to a person's inclination to explore beyond conventional boundaries in different aspects of life. Individuals with high levels of this personality trait tend to be more curious, creative and innovative in nature.

- You may try new things but would prefer not to venture too far beyond your comfort zone.
- You tend to be open to accepting abstract ideas after weighing them against existing solutions.
- You appreciate the arts to a certain extent but may lack the curiosity to explore them in depth.
- You may express your feelings only to people you are comfortable with.
- Your personality is more suited for jobs involving a mix of logical and creative thinking.

Emotional Stability

Emotional stability refers to the ability to withstand stress, handle adversity, and remain calm and composed when working through challenging situations. People with high levels of this personality trait tend to be more in control of their emotions and are likely to perform consistently despite difficult or unfavourable conditions.

- You are calm and composed in nature.
- You tend to maintain composure during high pressure situations.
- You are very confident and comfortable being yourself.
- You find it easy to resist temptations and practice moderation.
- You are likely to remain emotionally stable in jobs with high stress levels.

Polychronicity

Polychronicity refers to a person's inclination to multitask. It is the extent to which the person prefers to engage in more than one task at a time and believes that such an approach is highly productive. While this trait describes the personality disposition of a person to multitask, it does not gauge their ability to do so successfully.

- You prefer to work on one task at a time, complete it and then move on to the next.
- You prefer orderliness and likes to concentrate on the task at hand without any distractions.
- You can find it difficult to be placed in a work environment where there is a need to multitask or where expected to engage in multiple projects simultaneously.

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3 | Response

Automata

90 / 100 Code Replay

Question 1 (Language: Python3)

A company is transmitting its data to another server. To secure the data against malicious activity, they plan to reverse the data before transmitting. They want to know the number of data characters that do not change position even after the data stream is reversed. The network administrator has been tasked with ensuring the smooth transmission of the data.

Write an algorithm for the network administrator to help in finding the number of data characters that do not change position even after the data stream is reversed.

Scores

Programming Ability

80 / 100

Correct with inadvertent errors. Correct control structures and critical data dependencies incorporated. Some inadvertent errors make the code fail test cases.

Programming Practices

100 / 100

High readability, high on program structure. The source code is readable and does not consist of any significant redundant/improper coding constructs.

Functional Correctness

90 / 100

Correct basic and advanced functionality with partially correct corner case handling. Passes all the basic and advanced test cases in the test suite and a percentage of the edge test case(s).

Final Code Submitted **Compilation Status: Pass**

```

"""

"""
def unaffectedChar(dataStream):
    # Write your code here
    count =0
    n = len(dataStream)
    for i in range(n//2):
        if dataStream[i] == dataStream[n-1-i]:
            count+=2
    if n%2 !=0:
        count +=1

    return count

def main():
```

Code Analysis

Average-case Time Complexity

Candidate code: Complexity is reported only when the code is correct and it passes all the basic and advanced test cases.

Best case code: O(log N)

*N represents size of the string.

Errors/Warnings

There are no errors in the candidate's code.

Structural Vulnerabilites and Errors

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```
18    #input for dataStream
19    dataStream = str(input())
20
21
22    result = unaffectedChar(dataStream)
23    print(result)
24
25 if __name__ == "__main__":
26    main()
```

Readability & Language Best Practices

Line 8: Variable name doesn't conform to snake_case naming style

Test Case Execution

Passed TC: **90%**

Total score

9/10

100%	100%	50%
Basic(**4**/4)	Advance(**4**/4)	Edge(**1**/2)

Compilation Statistics

1	1	0	0	0	1
Total attempts	Successful	Compilation errors	Sample failed	Timed out	Runtime errors

Response time:	**00:03:30**
Average time taken between two compile attempts:	**00:03:30**
Average test case pass percentage per compile:	**90%**

Average-case Time Complexity

Average Case Time Complexity is the order of performance of the algorithm given a random set of inputs. This complexity is measured here using the Big-O asymptotic notation. This is the complexity detected by empirically fitting a curve to the run-time for different input sizes to the given code. It has been benchmarked across problems.

Test Case Execution

There are three types of test-cases for every coding problem:

Basic: The basic test-cases demonstrate the primary logic of the problem. They include the most common and obvious cases that an average candidate would consider while coding. They do not include those cases that need extra checks to be placed in the logic.

Advanced: The advanced test-cases contain pathological input conditions that would attempt to break the codes which have incorrect/semi-correct implementations of the correct logic or incorrect/semi-correct formulation of the logic.

Edge: The edge test-cases specifically confirm whether the code runs successfully even under extreme conditions of the domain of inputs and that all possible cases are covered by the code

Question 2 (Language: Python3)

There are N members in a marching band. Each member of the band has been assigned a value from 1 to N and they are arranged in ascending order of their assigned value. The band is participating in a competition in which

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they have to complete Q performances. After one particular performance, the two band members who were called out by the instructor had to swap positions with each other until all Q performances have been performed. Now the instructor wants to know the ID of the band member who is at the Kth position after Q performances.

Write an algorithm for the band instructor to find the ID of the band member who is at the Kth position after Q performances.

Scores

Programming Ability

80 / 100

Correct with inadvertent errors. Correct control structures and critical data dependencies incorporated. Some inadvertent errors make the code fail test cases.

Programming Practices

100 / 100

High readability, high on program structure. The source code is readable and does not consist of any significant redundant/improper coding constructs.

Functional Correctness

36 / 100

Partially correct basic functionality. The source code compiles and passes only some of the basic test cases. Some advanced or edge cases may randomly pass.

Final Code Submitted

Compilation Status: Pass

```
1
2 """
3
4 """
5 def bandMemberAtK(totalMembers, SwapMembers, posMe
  mber):
6   # Write your code here
7
8   arr=[]
9   for i in range(1,totalMembers+1):
10     arr.append(i)
11
12
13   for swap in SwapMembers:
14     i = swap[0]-1
15     k = swap[1]-1
16     for i in range(i,k//2):
17       temp = arr[i];
18       arr[i] = arr[k]
19       arr[k] = temp
20
21
22
23
24   return arr[posMember-1]
25
26
27 def main():
```

Code Analysis

Average-case Time Complexity

Candidate code: Complexity is reported only when the code is correct and it passes all the basic and advanced test cases.

Best case code: O(N)

*N represents number of members/performances.

Errors/Warnings

There are no errors in the candidate's code.

Structural Vulnerabilites and Errors

Readability & Language Best Practices

Line 13,24: too many blank lines (2)
Line 17: Unnecessary semicolon

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```
  #input for totalMembers
  totalMembers = int(input())

  #input for SwapMembers
  SwapMembers = []
  SwapMembers_rows,SwapMembers_cols  = map(int, input().split())
  for idx in range(SwapMembers_rows):
     SwapMembers.append(list(map(int, input().split())))

  #input for posMember
  posMember = int(input())


  result = bandMemberAtK(totalMembers, SwapMembers, posMember)
  print(result)

if __name__ == "__main__":
  main()
```

Test Case Execution

Passed TC: **66.67%**

Total score

8/12

80%	40%	100%
Basic(**4**/5)	Advance(**2**/5)	Edge(**2**/2)

Compilation Statistics

14	12	2	0	0	10
Total attempts	Successful	Compilation errors	Sample failed	Timed out	Runtime errors

Response time:	**00:32:56**
Average time taken between two compile attempts:	**00:02:21**
Average test case pass percentage per compile:	**12.5%**

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Average-case Time Complexity

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Edge: The edge test-cases specifically confirm whether the code runs successfully even under extreme conditions of the domain of inputs and that all possible cases are covered by the code

Automata Fix

58 / 100 Code Replay

Question 1 (Language: C++)

The function/method ***patternPrint*** accepts an argument *num*, an integer.
The function/method ***patternPrint*** prints *num* lines in the following pattern.

For example, *num* = 4, the pattern should be:
1
1 1
1 1 1
1 1 1 1

The function/method ***patternPrint*** compiles successfully but fails to print the desired result for some test cases due to incorrect implementation of the function/method. Your task is to fix the code so that it passes all the test cases.

Scores

Final Code Submitted — Compilation Status: Pass

```
// You can print the values to stdout for debugging
#include<bits/stdc++.h>
using namespace std;
void patternPrint(int num)
{
  int print=1;
  for(int i=0;i<num;i++)
  {
    for(int j=0;j<=i;j++)
    {
      cout<<print<<" ";
```

Code Analysis

Average-case Time Complexity

Candidate code: Complexity is reported only when the code is correct and it passes all the basic and advanced test cases.

Best case code:

*N represents

Errors/Warnings

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```
    }
    cout<<"\n";
  }
}

```

There are no errors in the candidate's code.

Structural Vulnerabilites and Errors

There are no errors in the candidate's code.

Test Case Execution

Passed TC: **100%**

Total score

8/8

100%	0%	100%
Basic(**7**/7)	Advance(**0**/0)	Edge(**1**/1)

Compilation Statistics

1	1	0	0	0	0
Total attempts	Successful	Compilation errors	Sample failed	Timed out	Runtime errors

Response time:	**00:00:48**
Average time taken between two compile attempts:	**00:00:48**
Average test case pass percentage per compile:	**100%**

Average-case Time Complexity

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Test Case Execution

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Edge: The edge test-cases specifically confirm whether the code runs successfully even under extreme conditions of the domain of inputs and that all possible cases are covered by the code

Question 2 (Language: C++)

The function/method ***allExponent*** returns a real number representing the result of exponentiation of base raised to power exponent for all input values. The function/method ***allExponent*** accepts two arguments - *baseValue*, an integer representing the base and *exponentValue*, an integer representing the exponent.

The incomplete code in the function/method ***allExponent*** works only for positive values of the exponent. You must

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complete the code and make it work for negative values of exponent as well.

Another function/method ***positiveExponent*** uses an efficient way for exponentiation but accepts only positive *exponent* values. You are supposed to use this function/method to complete the code in ***allExponent*** function/method.

Helper Description

The following function is used to represent a positiveExponent and is already implemented in the default code (Do not write this definition again in your code):

```
int positiveExponent(int baseValue, int exponentValue)

{

  /*It calculates the Exponent for the positive value of exponentValue

    This can be called as -

    int res = (float)positiveExponent(baseValue, exponentValue);*/

}
```

Scores

Final Code Submitted **Compilation Status: Pass**

```
// You can print the values to stdout for debugging
#include<bits/stdc++.h>
using namespace std;
float allExponent(int baseValue, int exponentValue)
{
  float res = 1;
  if(exponentValue >=0)
  {
    res = (float)positiveExponent(baseValue, exponentValue);
  }
  else
  {
    // write your code here for negative exponentInput
  }
  return res;
}


```

Code Analysis

Average-case Time Complexity

Candidate code: Complexity is reported only when the code is correct and it passes all the basic and advanced test cases.

Best case code:

*N represents

Errors/Warnings

There are no errors in the candidate's code.

Structural Vulnerabilites and Errors

There are no errors in the candidate's code.

Test Case Execution Passed TC: **66.67%**

Total score 4/6

100%	33%	100%
Basic(**2**/2)	Advance(**1**/3)	Edge(**1**/1)

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Compilation Statistics

2	1	1	1	0	0
Total attempts	Successful	Compilation errors	Sample failed	Timed out	Runtime errors

Response time:	**00:05:14**
Average time taken between two compile attempts:	**00:02:37**
Average test case pass percentage per compile:	**16.7%**

Average-case Time Complexity

Average Case Time Complexity is the order of performance of the algorithm given a random set of inputs. This complexity is measured here using the Big-O asymptotic notation. This is the complexity detected by empirically fitting a curve to the run-time for different input sizes to the given code. It has been benchmarked across problems.

Test Case Execution

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Edge: The edge test-cases specifically confirm whether the code runs successfully even under extreme conditions of the domain of inputs and that all possible cases are covered by the code

Question 3 (Language: C++)

Lisa always forgets her birthday which is on the 5th of July. So, develop a function/method which will be helpful to remember her birthday.

The function/method ***checkBirthDay*** return an integer '1' if it is her birthday else returns 0. The function/method ***checkBirthDay*** accepts two arguments - *month*, a string representing the month of her birthday and *day*, an integer representing the date of her birthday.

The function/method ***checkBirthDay*** compiles successfully but fails to return the desired result for some test cases. Your task is to fix the code so that it passes all the test cases.

Scores

Final Code Submitted **Compilation Status: Pass**

```
// You can print the values to stdout for debugging
#include<string.h>
```

Code Analysis

Average-case Time Complexity

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```
using namespace std;
int checkBirthDay(char* month, int day)
{
   if((strcmp(month,"July")) && (day==5))
         return 1;
      else
         return 0;
}

```

Candidate code: Complexity is reported only when the code is correct and it passes all the basic and advanced test cases.

Best case code:

*N represents

Errors/Warnings

There are no errors in the candidate's code.

Structural Vulnerabilites and Errors

There are no errors in the candidate's code.

Test Case Execution

Passed TC: **66.67%**

Total score

6/9

67%	67%	0%
Basic(**4**/6)	Advance(**2**/3)	Edge(**0**/0)

Compilation Statistics

2	2	0	1	0	0
Total attempts	Successful	Compilation errors	Sample failed	Timed out	Runtime errors

Response time:	**00:04:20**
Average time taken between two compile attempts:	**00:02:10**
Average test case pass percentage per compile:	**38.9%**

Average-case Time Complexity

Average Case Time Complexity is the order of performance of the algorithm given a random set of inputs. This complexity is measured here using the Big-O asymptotic notation. This is the complexity detected by empirically fitting a curve to the run-time for different input sizes to the given code. It has been benchmarked across problems.

Test Case Execution

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Edge: The edge test-cases specifically confirm whether the code runs successfully even under extreme conditions of the domain of inputs and that all possible cases are covered by the code

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Question 4 (Language: C++)

The function/method ***descendingSortArray*** performs an in-place sort on the given input list which will be sorted in descending order.
The function/method ***descendingSortArray*** accepts two arguments - *len,* an integer representing the length of the input list and *arr*, a list of integers representing the input list, respectively.
5
The function/method ***descendingSortArray*** compiles successfully but fails to get the desired result for some test cases due to logical errors. Your task is to fix the code so that it passes all the test cases.

Scores

Final Code Submitted **Compilation Status: Pass**

```
// You can print the values to stdout for debugging
#include<bits/stdc++.h>
using namespace std;
void descendingSortArray(int len, int* arr)
{
  int i, j, temp;
  for(i=0; i<=len-1;i++)
  {
    for(j=i; j<len;j++)
    {
      temp = 0;
      if(arr[i]<arr[j])
      {
        temp=arr[i];
        arr[i]=arr[j];
        arr[j]=temp;
      }
    }
  }
}
```

Code Analysis

Average-case Time Complexity

Candidate code: Complexity is reported only when the code is correct and it passes all the basic and advanced test cases.

Best case code:

*N represents

Errors/Warnings

There are no errors in the candidate's code.

Structural Vulnerabilites and Errors

There are no errors in the candidate's code.

Test Case Execution Passed TC: **100%**

Total score

10/10

100%	100%	0%
Basic(**6**/6)	Advance(**4**/4)	Edge(**0**/0)

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Compilation Statistics

2	2	0	1	0	0
Total attempts	Successful	Compilation errors	Sample failed	Timed out	Runtime errors

Response time:	**00:01:17**
Average time taken between two compile attempts:	**00:00:39**
Average test case pass percentage per compile:	**55%**

Average-case Time Complexity

Average Case Time Complexity is the order of performance of the algorithm given a random set of inputs. This complexity is measured here using the Big-O asymptotic notation. This is the complexity detected by empirically fitting a curve to the run-time for different input sizes to the given code. It has been benchmarked across problems.

Test Case Execution

There are three types of test-cases for every coding problem:

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Edge: The edge test-cases specifically confirm whether the code runs successfully even under extreme conditions of the domain of inputs and that all possible cases are covered by the code

Question 5 (Language: C++)

The function/method ***matrixSum*** returns an integer representing the sum of elements of the input matrix. The function/method ***matrixSum*** accepts three arguments - *rows*, an integer representing the number of rows of the input matrix, *columns*, an integer representing the number of columns of the input matrix and *matrix*, a two-dimensional array representing the input matrix.

The function/method ***matrixSum*** compiles unsuccessfully due to syntactical error. Your task is to debug the program so that it passes all test cases.

Scores

Final Code Submitted **Compilation Status: Pass**

```
// You can print the values to stdout for debugging
using namespace std;
int matrixSum(int rows, int columns, int **matrix)
{
```

Code Analysis

Average-case Time Complexity

Candidate code: Complexity is reported only when the code is correct and it passes all the basic and advanced

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```
int i, j, sum=0;
for(i=0;i<rows;i++)
{
  for(j=0;j<columns;j++)
    sum += matrix[i][j];
}
return sum;
}
```

test cases.

Best case code:

*N represents

Errors/Warnings

There are no errors in the candidate's code.

Structural Vulnerabilites and Errors

There are no errors in the candidate's code.

Test Case Execution

Passed TC: **100%**

Total score

10/10

100%	100%	0%
Basic(**6**/6)	Advance(**4**/4)	Edge(**0**/0)

Compilation Statistics

1	1	0	0	0	0
Total attempts	Successful	Compilation errors	Sample failed	Timed out	Runtime errors

Response time:	**00:00:27**
Average time taken between two compile attempts:	**00:00:27**
Average test case pass percentage per compile:	**100%**

Average-case Time Complexity

Average Case Time Complexity is the order of performance of the algorithm given a random set of inputs. This complexity is measured here using the Big-O asymptotic notation. This is the complexity detected by empirically fitting a curve to the run-time for different input sizes to the given code. It has been benchmarked across problems.

Test Case Execution

There are three types of test-cases for every coding problem:

Basic: The basic test-cases demonstrate the primary logic of the problem. They include the most common and obvious cases that an average candidate would consider while coding. They do not include those cases that need extra checks to be placed in the logic.

Advanced: The advanced test-cases contain pathological input conditions that would attempt to break the codes which have incorrect/semi-correct implementations of the correct logic or incorrect/semi-correct formulation of the logic.

Edge: The edge test-cases specifically confirm whether the code runs successfully even under extreme conditions of the domain of inputs and that all possible cases are covered by the code

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Question 6 (Language: C++)

The function/method ***selectionSortArray*** performs an in-place selection sort on the given input list which will be sorted in ascending order.
The function/method ***selectionSortArray*** accepts two arguments - *len,* an integer representing the length of the input list and *arr*, a list of integers representing the input list, respectively.

The function/method ***selectionSortArray*** compiles successfully but fails to get the desired result for some test cases due to logical errors. Your task is to fix the code so that it passes all the test cases.

Note:
In this particular implementation of selection sort, the smallest element in the list is swapped with the element at first index, the next smallest element is swapped with the element at the next index and so on.

Scores

Final Code Submitted **Compilation Status: Pass**

```
// You can print the values to stdout for debugging
void selectionSortArray(int len, int* arr){
  int x=0, y =0;
  for(x=0; x<len; x++){
    int index_of_min = x;
    for(y=x; y<len; y++){
      if(arr[index_of_min]>arr[y]){
        index_of_min = y;
      }
    }
    int temp = arr[x];
    arr[x] = arr[index_of_min];
    arr[index_of_min] = temp;
  }
}
```

Code Analysis

Average-case Time Complexity

Candidate code: Complexity is reported only when the code is correct and it passes all the basic and advanced test cases.

Best case code:

*N represents

Errors/Warnings

There are no errors in the candidate's code.

Structural Vulnerabilites and Errors

There are no errors in the candidate's code.

Test Case Execution Passed TC: **100%**

Total score 9/9

100%	100%	0%
Basic(**4**/4)	Advance(**5**/5)	Edge(**0**/0)

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Compilation Statistics

1	1	0	0	0	0
Total attempts	Successful	Compilation errors	Sample failed	Timed out	Runtime errors

Response time:	**00:00:40**
Average time taken between two compile attempts:	**00:00:40**
Average test case pass percentage per compile:	**100%**

Average-case Time Complexity

Average Case Time Complexity is the order of performance of the algorithm given a random set of inputs. This complexity is measured here using the Big-O asymptotic notation. This is the complexity detected by empirically fitting a curve to the run-time for different input sizes to the given code. It has been benchmarked across problems.

Test Case Execution

There are three types of test-cases for every coding problem:

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Edge: The edge test-cases specifically confirm whether the code runs successfully even under extreme conditions of the domain of inputs and that all possible cases are covered by the code

Question 7 (Language: C++)

You are given a predefined structure Point and also a collection of related functions/methods that can be used to perform some basic operations on the structure.

You must implement the function/method ***isTriangle*** which accepts three points *P1, P2, P3* as inputs and checks whether the given three points form a triangle.

If they form a triangle, the function/method returns an integer 1. Otherwise, it returns an integer 0.

.

Helper Description
The following class is used to represent point and is already implemented in the default code (Do not write these definitions again in your code):

```
class Point

{
```

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```
    private:

    int X;

    int Y;

    double Point_calculateDistance(Point *point1, Point *point2)

    {

        /*Return the euclidean distance between two input points.

        This can be called as -

        *  If P1 and P2 are two points then -

        *  P1->Point_calculateDistance(P2);*/

     }

}
```

Scores

Final Code Submitted

Compilation Status: Fail

```
1  // You can print the values to stdout for debugging
2  using namespace std;
3  int isTriangle(Point *P1, Point *P2, Point *P3)
4  {
5     // write your code here
6     if(P1<0 && P2 <0 && P3){
7
8     }
9
10 }
11
```

Code Analysis

Average-case Time Complexity

Candidate code: Complexity is reported only when the code is correct and it passes all the basic and advanced test cases.

Best case code:

*N represents

Errors/Warnings

```
In file included from main_25.cpp:8:
source_25.cpp: In function 'int isTriangle(Point*, Point*, Point*)':
source_25.cpp:10:1: error: no return statement in function returning non-void [-Werror=return-type]
}
^
cc1plus: some warnings being treated as errors
```

Structural Vulnerabilites and Errors

There are no errors in the candidate's code.

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Compilation Statistics

Total attempts	Successful	Compilation errors	Sample failed	Timed out	Runtime errors
0	0	0	0	0	0

Response time:	**00:06:39**
Average time taken between two compile attempts:	**00:00:00**
Average test case pass percentage per compile:	**0%**

Average-case Time Complexity

Average Case Time Complexity is the order of performance of the algorithm given a random set of inputs. This complexity is measured here using the Big-O asymptotic notation. This is the complexity detected by empirically fitting a curve to the run-time for different input sizes to the given code. It has been benchmarked across problems.

Test Case Execution

There are three types of test-cases for every coding problem:

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Edge: The edge test-cases specifically confirm whether the code runs successfully even under extreme conditions of the domain of inputs and that all possible cases are covered by the code

WriteX - Essay Writing

75 / 100 | CEFR: **C1**

Question

Some parents feel that sports is a distraction to their kids' studies. There are others who give due importance to sports for the holistic development of a child.
What is your view? Support your response with reasons and examples.

Scores

Content Score	Grammar Score
81 / 100	**59** / 100

Response

Many Parents worry that sports distract their children from academic success , while others believe sports are essential for a child overall development . Both perspective s have valid argument and understanding them helps create a balanced approach to education. On one hand , some parents prioritize academic achievement abov all els

Error Summary

Spelling	**37**

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e ,In today's compitetive world securing admission to prestigious institutions such as Harvard university or IIIT requires excellent grades and consistent academic perform ance . These parents fear that spending time on sports may reduce study hour , lead ing lower marks , They argue that children already face heavy academic workloads, i nluding homework , projects and exam preparatins. Adding regular sports parctice c ould result in fatigue and reduce concentraion . Moreover , nowt evry child becomes a professional athelete , so some parents see sports as as risky investment of time t hat may not directly contribute to future carrer stability. On the other hand , many p arents strongly supported sports as a vital component of holistic development . They believe education is not limited to text book and examination . Sports help children d evelop physical fitness , mental strength , discipline , teamwork and leadership skills . For example globally admired athletes like virat kohli and rohit sharma how dedicat ion to sports can build resillence ,confidence ,a nd a strong work ethic . Even for chil dren who do not persue sports pofessionaly , participation in games teaches valuable life lessons such as handling success and failure gracefully . Furthermore , research has shown that physcially activity improve brain function , memory , and cocentratio n . Chldren who engage n sport often manage tei time better and perform well acade mially . Sports also reduce stress and anxiety , which are increasingly common amo ng students . In addition , In Playkg aports is also good for mental stress gave good sleep at night , make muscle stronfg . It also encorage social interaction , helping chi ldren build freindship and communication skills . These Qualities contribute significall y to emotional and social development . In conclusion , while cncerns about sports af fecting academic are understandle , completely restricting children from sports may l imit their overall growth . A balanced approach is essential . Parents should encoura ge their children to focus on studies while also particiaptig in sports in as tructured a nd oderate manner . When managed properly sports do not ditract kid from a educat ion.

White Space	43
Style	0
Grammar	20
Typographical	0

Essay Statistics

383	21	18	247	90
Total words	Total sentences	Average sentence length	Total unique words	Total stop words

Error Details

Spelling

...parents prioritize academic achievement abov all else , In today's compitetive world ...	Possible spelling mistake found
...c achievement abov all else ,In today's compitetive wo rld securing admission to prestigious...	Possible spelling mistake found
...titutions such as Harvard university or IIIT requires ex cellent grades and consisten...	Possible spelling mistake found
... already face heavy academic workloads, inluding hom ework , projects and exam preparatin...	Possible spelling mistake found
..., inluding homework , projects and exam preparatins. Adding regular sports parctice could r...	Possible spelling mistake found
...exam preparatins. Adding regular sports parctice could result in fatigue and reduce conc...	Possible spelling mistake found

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...tice could result in fatigue and reduce concentraion . M oreover , nowt evry child becomes a ...	Possible spelling mistake found
...d reduce concentraion . Moreover , nowt evry child bec omes a professional athelete ,...	Possible spelling mistake found
... nowt evry child becomes a professional athelete , so s ome parents see sports as as risk...	Possible spelling mistake found
...t may not directly contribute to future carrer stability. On the other hand , many par...	Possible spelling mistake found
education is not limited to text book and examination . S ports	Possible spelling mistake found. Consider replacing the highlighted text with: 'textbook'.
... example globally admired athletes like virat kohli and rohit sharma how dedication t...	Possible spelling mistake found
...le globally admired athletes like virat kohli and rohit sh arma how dedication to spor...	Possible spelling mistake found
...y admired athletes like virat kohli and rohit sharma ho w dedication to sports can bui...	Possible spelling mistake found
...red athletes like virat kohli and rohit sharma how dedic ation to sports can build resi...	Possible spelling mistake found
...arma how dedication to sports can build resillence ,con fidence ,a nd a strong work ethic ...	Possible spelling mistake found
...ts can build resillence ,confidence ,a nd a strong work ethic . Even for children...	Possible spelling mistake found
...rk ethic . Even for children who do not persue sports p ofessionaly , participation in ...	Possible spelling mistake found
...n for children who do not persue sports pofessionaly , participation in games teaches valua...	Possible spelling mistake found
.... Furthermore , research has shown that physcially acti vity improve brain function , memor...	Possible spelling mistake found
...y improve brain function , memory , and cocentration . Chldren who engage n sport often mana...	Possible spelling mistake found
... function , memory , and cocentration . Chldren who e ngage n sport often manage tei tim...	Possible spelling mistake found
cocentration . Chldren who engage n sport often manage tei	Possible spelling mistake found. Consider replacing the highlighted text with: 'in'.
...Chldren who engage n sport often manage tei time bet ter and perform well academiall...	Possible spelling mistake found
...manage tei time better and perform well academially . Sports also reduce stress and anxiety...	Possible spelling mistake found
...ommon among students . In addition , In Playkg aport s is also good for mental stress g...	Possible spelling mistake found
...mong students . In addition , In Playkg aports is also g ood for mental stress gave goo...	Possible spelling mistake found
... gave good sleep at night , make muscle stronfg . It al so encorage social interaction ,...	Possible spelling mistake found
...al interaction , helping children build freindship and co mmunication skills . These Qualit...	Possible spelling mistake found
...ion skills . These Qualities contribute significally to em otional and social development . I...	Possible spelling mistake found

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...ial development . In conclusion , while cncerns about s ports affecting academic are und...	Possible spelling mistake found
...rns about sports affecting academic are understandle , completely restricting children from ...	Possible spelling mistake found
...children to focus on studies while also particiaptig in sp orts in as tructured and oderate m...	Possible spelling mistake found
...while also particiaptig in sports in as tructured and ode rate manner . When managed prope...	Possible spelling mistake found
...ticiaptig in sports in as tructured and oderate manner . When managed properly sports d...	Possible spelling mistake found
...r . When managed properly sports do not ditract kid fr om a education.	Possible spelling mistake found
...properly sports do not ditract kid from a education.	Use "an" instead of 'a' if the following word starts with a vowel sound, e.g. 'an article', 'an hour'

White Space

...Parents worry that sports distract their children from a cademic success , while o...	Possible typo: you repeated a whitespace
...ct their children from academic success , while others believe sports are essenti...	Put a space after the comma, but not before the comma
...ssential for a child overall development . Both perspect ives have valid argument a...	Don't put a space before the full stop
...anced approach to education. On one hand , some par ents prioritize academic achiev...	Put a space after the comma, but not before the comma
...itize academic achievement abov all else ,In today's co mpitetive world securing ad...	Put a space after the comma, but not before the comma
...ades and consistent academic performance . These par ents fear that spending time o...	Don't put a space before the full stop
...ing time on sports may reduce study hour , leading lo wer marks , They argue that c...	Put a space after the comma, but not before the comma
... reduce study hour , leading lower marks , They argue that children already face h...	Put a space after the comma, but not before the comma
...vy academic workloads, inluding homework , projects and exam preparatins. Adding r...	Put a space after the comma, but not before the comma
...esult in fatigue and reduce concentraion . Moreover , n owt evry child becomes a pr...	Don't put a space before the full stop
...tigue and reduce concentraion . Moreover , nowt evry child becomes a professional ...	Put a space after the comma, but not before the comma
...ry child becomes a professional athelete , so some par ents see sports as as risky ...	Put a space after the comma, but not before the comma
...ture carrer stability. On the other hand , many parents strongly supported sports ...	Put a space after the comma, but not before the comma
... vital component of holistic development . They believe education is not limited t...	Don't put a space before the full stop
...not limited to text book and examination . Sports help children develop physical f...	Don't put a space before the full stop

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...s help children develop physical fitness , mental streng th , discipline , teamwork...	Put a space after the comma, but not before the comma
...velop physical fitness , mental strength , discipline , te amwork and leadership sk...	Put a space after the comma, but not before the comma
...l fitness , mental strength , discipline , teamwork and l eadership skills . For ex...	Put a space after the comma, but not before the comma
...cipline , teamwork and leadership skills . For example globally admired athletes l...	Don't put a space before the full stop
...edication to sports can build resillence ,confidence ,a n d a strong work ethic	Put a space after the comma, but not before the comma
...sports can build resillence ,confidence ,a nd a strong w ork ethic . Even for chil...	Put a space after the comma, but not before the comma
...e ,confidence ,a nd a strong work ethic . Even for child ren who do not persue spo...	Don't put a space before the full stop
...en who do not persue sports pofessionaly , participatio n in games teaches valuabl...	Put a space after the comma, but not before the comma
...ts pofessionaly , participation in games teaches valuabl e life lessons such as ha...	Possible typo: you repeated a whitespace
... handling success and failure gracefully . Furthermore , research has shown that p...	Don't put a space before the full stop
...ess and failure gracefully . Furthermore , research has shown that physcially acti...	Put a space after the comma, but not before the comma
...yscially activity improve brain function , memory , and cocentration . Chldren who...	Put a space after the comma, but not before the comma
...activity improve brain function , memory , and cocentr ation . Chldren who engage n...	Put a space after the comma, but not before the comma
...ain function , memory , and cocentration . Chldren wh o engage n sport often manage...	Don't put a space before the full stop
...time better and perform well academially . Sports also reduce stress and anxiety ,...	Don't put a space before the full stop
... . Sports also reduce stress and anxiety , which are inc reasingly common among stu...	Put a space after the comma, but not before the comma
...h are increasingly common among students . In additio n , In Playkg aports is also ...	Don't put a space before the full stop
...ngly common among students . In addition , In Playkg aports is also good for menta...	Put a space after the comma, but not before the comma
...r mental stress gave good sleep at night , make muscl e stronfg . It also encorage ...	Put a space after the comma, but not before the comma
...ood sleep at night , make muscle stronfg . It also enco rage social interaction , h...	Don't put a space before the full stop
...fg . It also encorage social interaction , helping childre n build freindship and c...	Put a space after the comma, but not before the comma
...uild freindship and communication skills . These Qualiti es contribute significally...	Don't put a space before the full stop
...ally to emotional and social development . In conclusio n , while cncerns about spo...	Don't put a space before the full stop
...l and social development . In conclusion , while cncern s about sports affecting ac...	Put a space after the comma, but not before the comma

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...orts affecting academic are understandle , completely r estricting children from sp...	Put a space after the comma, but not before the comma
...om sports may limit their overall growth . A balanced a pproach is essential . Pare...	Don't put a space before the full stop
...rowth . A balanced approach is essential . Parents shou ld encourage their children...	Don't put a space before the full stop
...ports in as tructured and oderate manner . When man aged properly sports do not dit...	Don't put a space before the full stop

Grammar

Many Parents worry that sports distract their children fro m academic success , while others believe sports are ess ential for a child overall development .	Possible grammar error found. Consider inserting "is" over here.
Both perspectives have valid argument and understandin g them helps create a balanced approach to education.	Possible grammar error found. Consider replacing it with "arguments".
On one hand , some parents prioritize academic achieve ment abov all else ,In today's compitetive world securing admission to prestigious institutions such as Harvard uni versity or IIIT requires excellent grades and consistent a cademic performance .	Possible grammar error found. Consider inserting "the" over here.
On one hand , some parents prioritize academic achieve ment abov all else ,In today's compitetive world securing admission to prestigious institutions such as Harvard uni versity or IIIT requires excellent grades and consistent a cademic performance .	Possible grammar error found. Consider replacing it with "world,".
These parents fear that spending time on sports may red uce study hour , leading lower marks , They argue that c hildren already face heavy academic workloads, inluding homework , projects and exam preparatins.	Possible grammar error found. Consider replacing it with "hours,".
These parents fear that spending time on sports may red uce study hour , leading lower marks , They argue that c hildren already face heavy academic workloads, inluding homework , projects and exam preparatins.	Possible grammar error found. Consider inserting "to" over here.
These parents fear that spending time on sports may red uce study hour , leading lower marks , They argue that c hildren already face heavy academic workloads, inluding homework , projects and exam preparatins.	Possible grammar error found. Consider replacing it with "marks.".
Moreover , nowt evry child becomes a professional athele te , so some parents see sports as as risky investment of time that may not directly contribute to future carrer sta bility.	Possible grammar error found. Consider removing "nowt" from here.
On the other hand , many parents strongly supported sp orts as a vital component of holistic development .	Possible grammar error found. Consider replacing it with "support".
They believe education is not limited to text book and ex amination .	Possible grammar error found. Consider replacing it with "books".
They believe education is not limited to text book and ex amination .	Possible grammar error found. Consider replacing it with "examinations.".
For example globally admired athletes like virat kohli and rohit sharma how dedication to sports can build resillenc e ,confidence ,a nd a strong work ethic .	Possible grammar error found. Consider replacing it with "example,".

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Even for children who do not persue sports pofessionaly , participation in games teaches valuable life lessons such as handling success and failure gracefully .	Possible grammar error found. Consider replacing it with "sports,".
Furthermore , research has shown that physcially activity improve brain function , memory , and cocentration .	Possible grammar error found. Consider replacing it with "improves".
In addition , In Playkg aports is also good for mental stre ss gave good sleep at night , make muscle stronfg .	Possible grammar error found. Consider removing "In" from here.
In addition , In Playkg aports is also good for mental stre ss gave good sleep at night , make muscle stronfg .	Possible grammar error found. Consider removing "gave" from here.
In addition , In Playkg aports is also good for mental stre ss gave good sleep at night , make muscle stronfg .	Possible grammar error found. Consider replacing it with "makes".
In addition , In Playkg aports is also good for mental stre ss gave good sleep at night , make muscle stronfg .	Possible grammar error found. Consider replacing it with "muscles".
In conclusion , while cncerns about sports affecting acad emic are understandle , completely restricting children fr om sports may limit their overall growth .	Possible grammar error found. Consider replacing it with "academics".
When managed properly sports do not ditract kid from a education.	Possible grammar error found. Consider replacing it with "children".

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4 Learning Resources

English Comprehension

- Improve your knowledge of Business English
- Improve your hold on the language by reading Shakespearan plays
- Learn about how to get better at reading

Logical Ability

- Learn about how to solve problems by deriving complex rules
- Learn about syllogisms- the basics of deduction
- Practice examples of deductive reasoning

Quantitative Ability (Advanced)

- Learn about logarithms
- Learn about the concepts of probability and statistics
- Learn about permutations and combinations: from the shallows to the deep

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